

Mohammad Saifullah Khan

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Research Interests

Research Interests

Education

Indian Institute of Science Education and Research Bhopal, Electrical Engineering and Computer Science, minor in Data Science Bhopal, India
Dec 2021 – June 2026

Holy Family Convent Sr. Sec. School, Secondary Education Bhopal, India
May 2021

Experience

Multi-Robot Autonomy (MOON) Lab, IISER Bhopal, Project Associate I Bhopal, India
July 2026 – present
3 months

- Developing a bio-inspired surface vehicle for autonomous navigation in vegetation-choked and debris-laden waterways, where conventional propellers foul.
- Building a Unity-based simulation environment for the lab's marine, ground and aerial platforms.
- Investigating sim-to-real transfer of learned navigation policies onto the lab's ASV fleet, addressing hydrodynamic mismatch, sensor noise, actuation delay, etc.

Multi-Robot Autonomy (MOON) Lab, IISER Bhopal, Graduate Researcher Bhopal, India
June 2025 – June 2026
1 year 1 month

- Developed MA-BaTS, a multi-agent adaptive bathymetric mapping framework using Gaussian Process field reconstruction with Thompson Sampling acquisition; achieved ~68% reduction in traversal distance at equivalent reconstruction error, validated on a two-USV team in outdoor field trials.
- Built and ran the full deployment stack for these trials with ROS coordination, echosounder and GPS/IMU integration, and mission execution on water.

Department of EEE, IIT Guwahati, Visiting Student Researcher (concurrent with MOON Lab) Guwahati, India
Mar 2025 – Oct 2025
8 months

- Visiting researcher at the Energy Resilience Research Lab under Dr. Prabir Barooah.
- Built an autonomous field-testing platform and conducted real-world experimental validation.

Multi-Robot Autonomy (MOON) Lab, IISER Bhopal, Undergraduate Researcher Bhopal, India
Aug 2023 – Apr 2025
1 year 9 months

- Co-developed a search-based collision avoidance strategy encoding COLREGs maritime traffic rules for autonomous surface vessels, enabling rule-compliant safe navigation.
- Prototyped autonomous robotic systems covering sensing, control, deployment, and field trials.

Projects

MA-BaTS: Multi-agent Adaptive Bathymetry Mapping using Thompson Sampling June 2025 – present
MOON Lab, IISER Bhopal

- Developed a learning-based adaptive framework for collaborative bathymetric mapping, combining Gaussian Process reconstruction on the seafloor depth field with multi-robot decision making and control under limited sensing.
- Achieved a ~68% reduction in traversal distance at equivalent reconstruction error; framework validated in real-world field deployments.

EcoBoat: Autonomous Cleaning Boat (v2) MOON Lab, IISER Bhopal - Developed an autonomous lake-cleaning robot with Tesla-valve-inspired passive debris collection. - Implemented random-sampling-based coverage planning for debris collection missions.	Dec 2024 – Feb 2025
AquaRP: Autonomous Surface Vehicle (ASV) Research Platform MOON Lab, IISER Bhopal & IIT Guwahati - Designed and integrated a low-cost (\$2200) autonomous surface vehicle with embedded control (Pixhawk), onboard computation (NVIDIA Jetson), and multi-modal sensing (camera, GPS/IMU, echosounder). - Implemented autonomous navigation and multi-agent coordination, enabling collaborative field missions and real-world system validation.	Aug 2024 – Sept 2024
Manipulator Arm for Pick-and-Place Task MOON Lab, IISER Bhopal - Designed a custom 5-DOF manipulator; implemented kinematic control using Robotics Toolbox. - Integrated vision-based object pose estimation using FoundationPose.	Dec 2023 – July 2024
Autonomous Cleaning Boat MOON Lab, IISER Bhopal - Developed an autonomous debris-cleaning robot with YOLOv5s-based detection in VRX simulation. - Deployed onboard computation using Jetson Nano with Pixhawk-based navigation.	May 2023 – July 2023
GesTr: Wearable Indian Sign Language Translation Device ED'ICT Lab, IISER Bhopal Developed a wearable ISL-to-text/audio system with 97% real-time accuracy; shortlisted at 2022-23 IEEE CASS SDC.	Dec 2022 – Mar 2023

Awards

Bharat Innovates 2026 (National Basecamp) Shortlisted from 1,400 teams to the top 400 to showcase GesTr.	Dec 2025
Best Oral Presentation (EECS Symposium) Presented a marine robotics platform for in-field experimentation and its application in water body cleaning.	Feb 2025
Inclusion@RSS 2024 Fellowship Awarded a fellowship to attend the Robotics: Science and Systems (RSS) 2024 Conference at TU Delft, the Netherlands; selected as one of 27 students globally.	July 2024
ISRO Robotics Challenge-URSC 2024 (IRoC-U2024) Qualified the Hardware Functionality Demonstration and Rover Functionality Demonstration rounds; selected as one of the top 20 teams in India.	July 2024
IASc-NSA-NASI Summer Research Fellowship Awarded a competitive national summer research fellowship to conduct summer research under Dr Nagahanumaiah at CMTI, Bengaluru.	May 2024
2022-23 IEEE CASS Student Design Competition Achieved third position in Asia-Pacific.	Mar 2023

Publications

A Search-Based Collision Avoidance Strategy with COLREGs for Autonomous Surface Vehicles

Conference paper (Oral Presentation).

K. Jaiswal, **M. S. Khan**, R. Kulkarni, S. P. Baliyarasimhuni

Patents

A User-Wearable Indian Sign Language Translator

Oct 2023

Application number: 202321071011; Date of filing: 18/10/2023.

Teaching and Outreach

Department of EECS, IISER Bhopal, Teaching Assistant

Bhopal, India

- Teaching assistant for ECS 203 (Basic Electronics); led tutorials and supported circuit analysis and lab concepts.

Aug 2024 – Nov 2024

Ashoka University, Invited Workshop Speaker

Sonipat, India

- Gave a hands-on introduction to robotics and ROS2 to undergraduate CS and Physics students.

Feb 2026 – Feb 2026

Skills

Programming

Robotics & Middleware

Embedded Systems & Hardware Integration

Perception & Sensors

Planning & Control

Simulation

Design & Fabrication

Machine Learning

Other Tools & Technologies

Languages

Hindi

Native speaker

English

Fluent

References

Dr Sujit Pedda Baliyarasimhuni

Dr Pydi Ganga Bahubalindrani